Module 2

Depth Estimation and Multi-Camera Views

Perspective • Stereo • Epipolar Geometry • Homography • Rectification • DLT • RANSAC • 3-D Reconstruction • Auto-Calibration

Depth Estimation

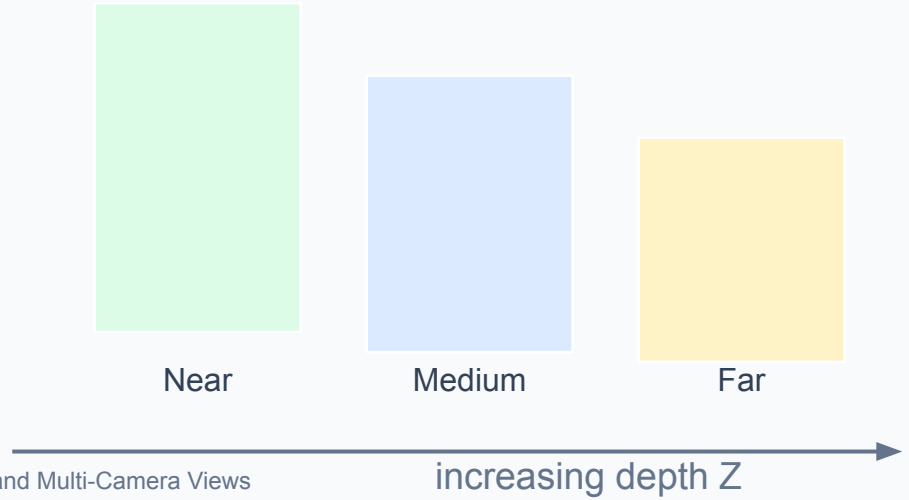
- Depth Estimation is a computer vision technique in which we measure the distance of an object in image or video from the camera
- Depth means distance from the camera to a scene point.
- In real , image appear as 2D

$$\boldsymbol{Image\ point}: \boldsymbol{p} = (\boldsymbol{u}, \boldsymbol{v})$$

- But in real world, image is 3D

$$\boldsymbol{3-D\ point}: \boldsymbol{P} = (\boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z})$$

- Thus the objective of depth estimation is to recover the missing Z dimension from images



RGB image tells WHAT is visible; depth tells WHERE it is in 3-D.

Depth Estimation

- Depth is extracted from either monocular (single) or stereo (multiple views of a scene) images.

  Using one camera

  A single image estimates depth using visual clues such as:

  - object size
  - overlapping objects
  - shadows and texture
  - perspective lines
  - trained deep-learning models

Depth Estimation

Using two cameras

Two cameras capture the same object from slightly different positions. The object appears at different locations in both images. This difference is called **disparity**.

$$z = f\frac{B}{d}$$

where:

Z = depth of the object

f = camera focal length

B = distance between the two cameras, called baseline

d = disparity

Large disparity → object is near

Small disparity → object is far

- Traditional methods use multi-view geometry to find the relationship between the images. Newer methods can directly estimate depth by minimizing the regression loss, or by learning to generate a novel view from a sequence.
- It is an important task in computer vision and has various applications such as 3D reconstruction, augmented reality, autonomous navigation, and more.
- **Depth Estimation techniques**

  Stereo vision: Uses two cameras. Depth is calculated from the difference in object position, called disparity.

  Monocular depth estimation: Uses one image and estimates depth from object size, texture, shadow, overlap, and perspective.

  Motion-based depth

  Uses multiple frames from a moving camera. Nearby objects appear to move faster than far objects.

  LiDAR: Sends laser pulses and measures the time taken for them to return.

  Time-of-Flight camera: Uses infrared light and measures its travel time to calculate distance.

  Structured light: Projects a known light pattern on the scene and measures its deformation.

  Deep-learning methods: Neural networks learn to predict a depth map from one or more images.

Depth Map

- Depth map is a image-like representation in which every pixel intensity represents the distance of that point from the camera.
- A depth map stores one depth value for each pixel
- Generally

$$Bright\ pixels \rightarrow near\ to\ camera$$

$$Dark\ pixels \rightarrow far\ from\ camera$$

- The output depth map will have the same size as of original image

Why Needed?

- Objects are visible in a camera image, but a computer does not naturally know how far away each object is.
- For example, suppose the camera of an autonomous car detects a pedestrian on the road. Detecting the pedestrian alone is not sufficient.
- The car must also determine:
  - Is the pedestrian 3 metres away or 30 metres away?
  - Is the pedestrian directly in front of the car or positioned to the side?
  - Should the brakes be applied immediately?
  - Is there another object behind the pedestrian?
  - Therefore, depth information is extremely important along with object detection.

Why Depth Is Ambiguous in a Single Image?

$$x = f\frac{}{Z}$$

$$y = f\frac{Y}{Z}$$

- where

  ((X, Y, Z)) represents a 3-D point in the real world.

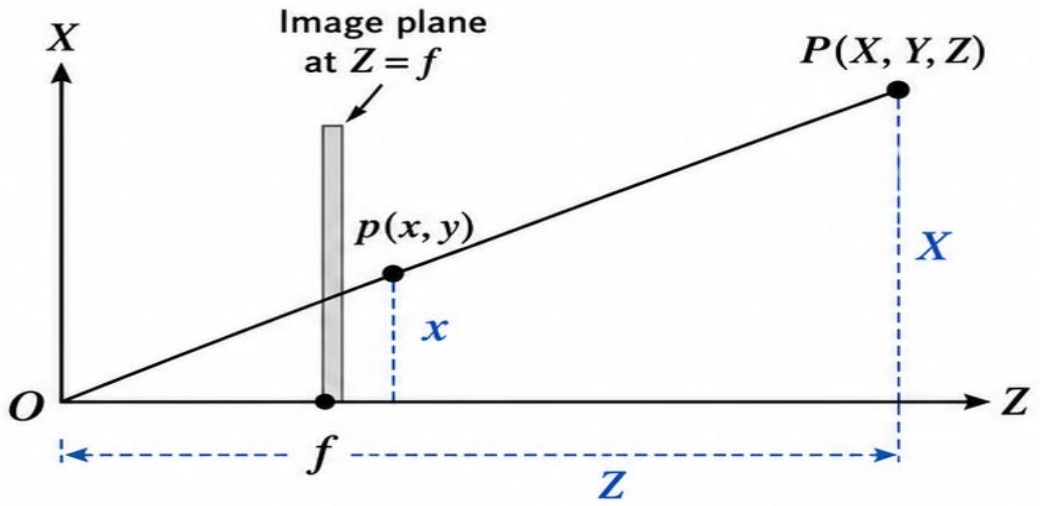
  ((x, y)) represents the corresponding point on the image plane.

  (f) is the focal length of the camera.

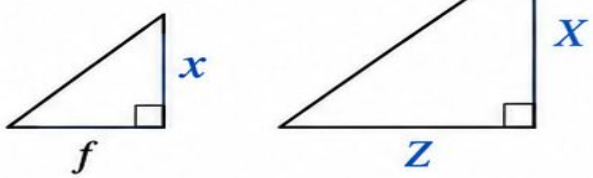
  (Z) represents the depth of the object from the camera.

- Since there are only two equations for three unknown variables, the exact 3-D position of the point

1) X–Z Plane (top/side view)

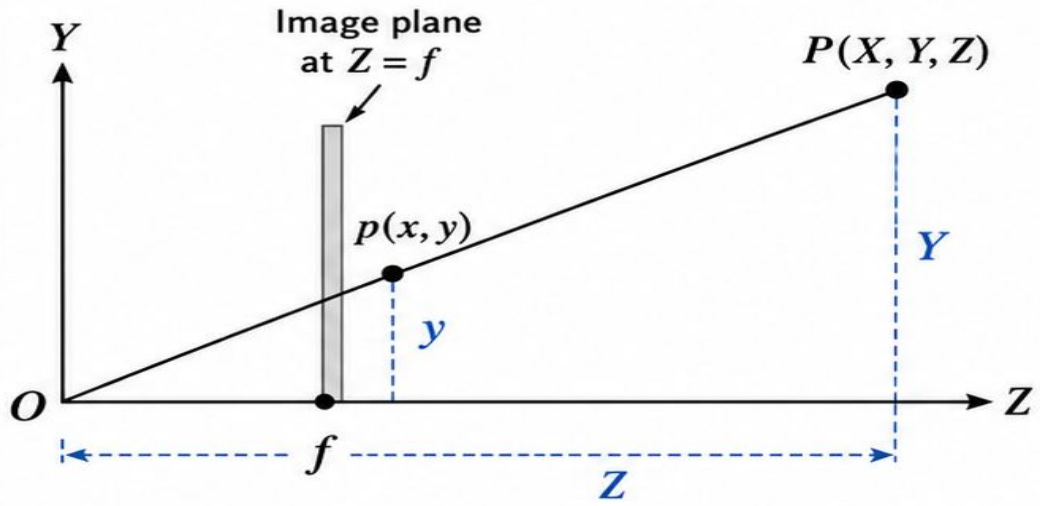


Similar triangles:

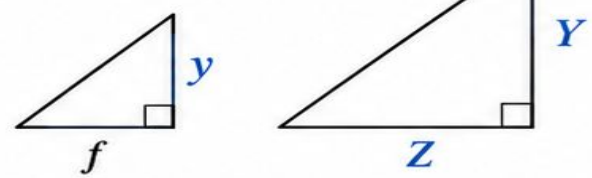


$$\frac{x}{f} = \frac{X}{Z} \quad \Rightarrow \quad x = \frac{fX}{Z}$$

2) Y–Z Plane (top/side view)



Similar triangles:



$$\frac{y}{f} = \frac{Y}{Z} \quad \Rightarrow \quad y = \frac{fY}{Z}$$

(X, Y, Z)	=	3-D world point
(x, y)	=	image-plane point
f	=	focal length
Z	=	depth

Perspective projection is derived using similar triangles in the pinhole camera model.

Size–Distance Ambiguity

- A small object placed close to the camera and a large object placed far away from the camera can appear to have the same size in an image.
- The approximate perspective relationship is

$$Image\ size \propto \frac{Actual\ object\ size}{Depth}$$

This means that the image size increases when the actual object size increases, but decreases when the object moves farther from the camera.

- Therefore, depth cannot be estimated reliably from image size alone unless the actual size of the object is already known. This problem is called size–distance ambiguity

Scale Ambiguity

- A single image can provide information about the **relative structure** of a scene, but the actual metric scale of the scene may remain unknown.
- For example Object B is approximately twice as far from the camera as Object A. However, the actual distances could be:

  2 m and 4 m

  5 m and 10 m

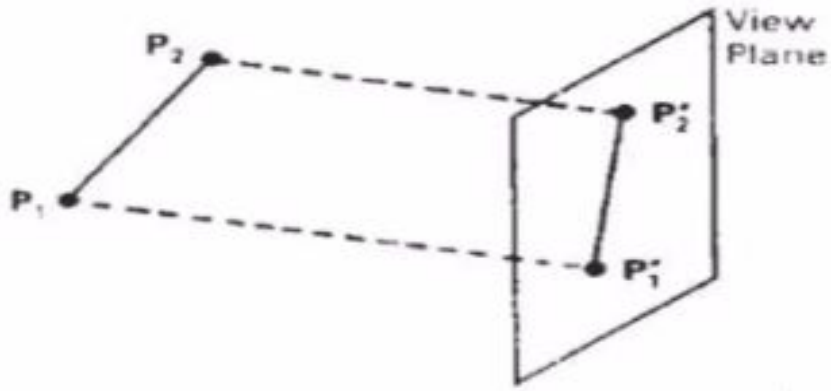
  20 m and 40 m This problem is known as **scale ambiguity**.

Occlusion Ambiguity

- It occurs when one object partially or completely blocks another object in an image.

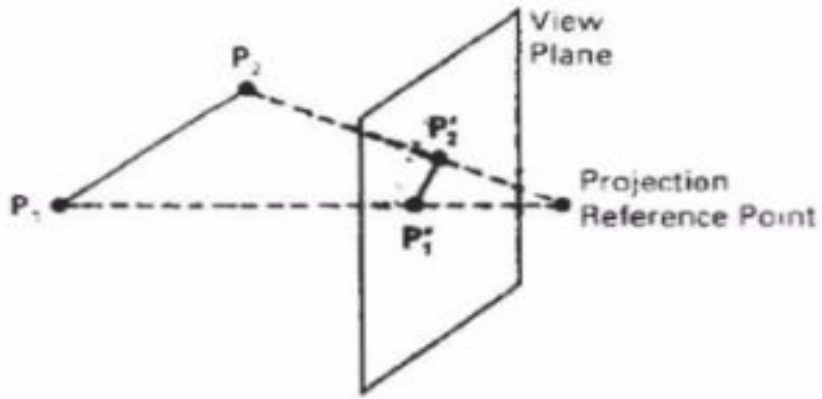
3D Projection

- Projection is a method of mapping a 3 dimensional object into a 2 dimensional view plane.
- Mapping is determined by the projection line called **Projector.**
- **"Projection can be defined as a mapping of point P(x, y, z) onto its image P`(x`, y`, z`) in the projection plane, which constitute the display surface."**



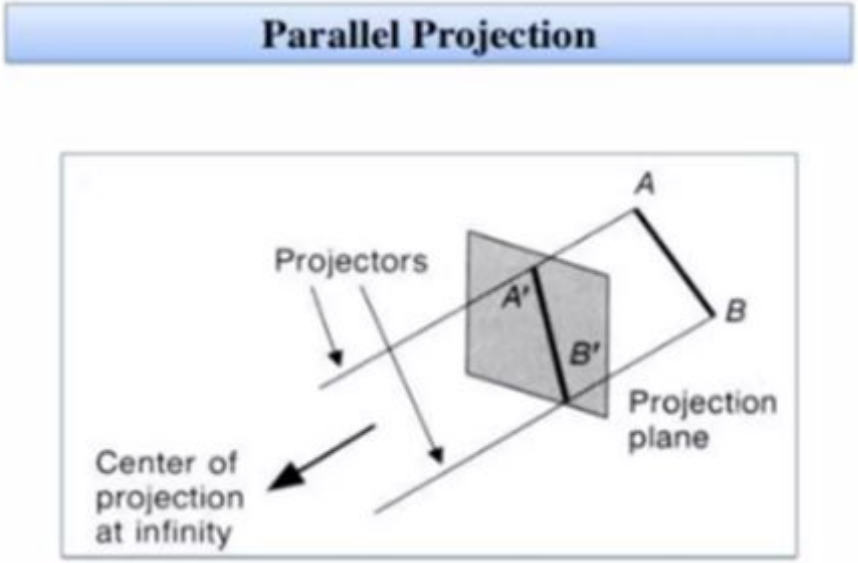
Types of projection

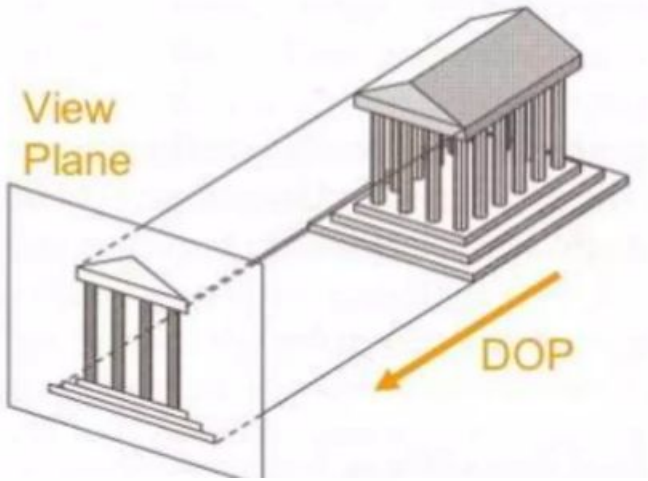
- Two types of projections are there.
- Those are **Perspective** projection and **Parallel** projection.
- **Projection Reference Point(PRP)** or Center of Projection point is a point, where projection lines converges.



Parallel projection

- In parallel projection, projection lines are parallel to each other.



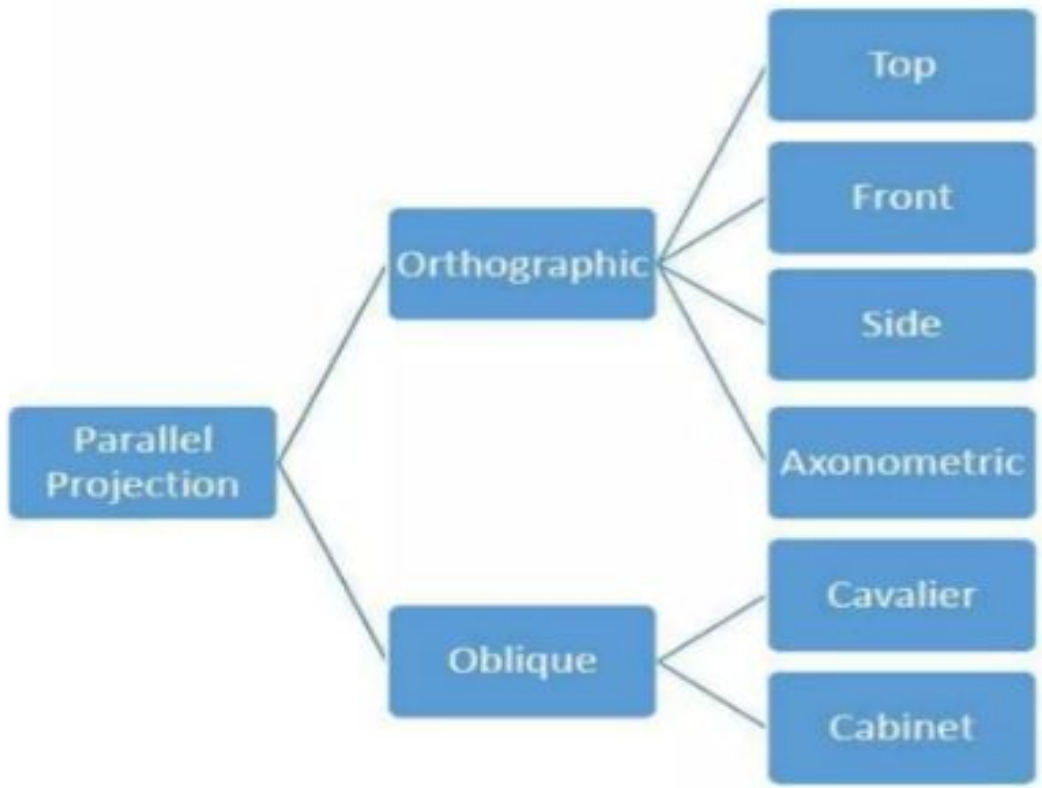


- Image on the view plane of the object will be equal size with the object.

Types of Parallel projection

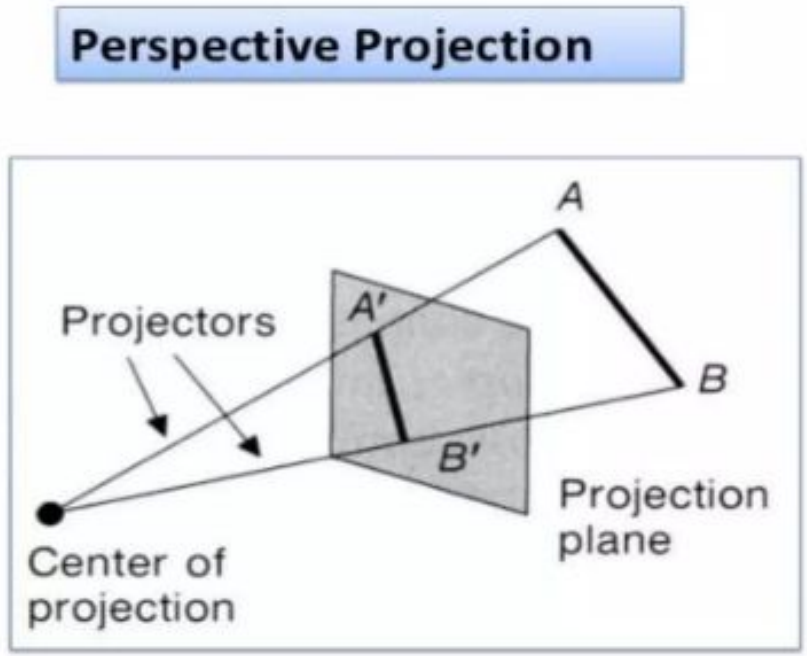
- **Projection Reference Point(PRP)** lies in infinite position.

- Different types of parallel projection are given bellow:



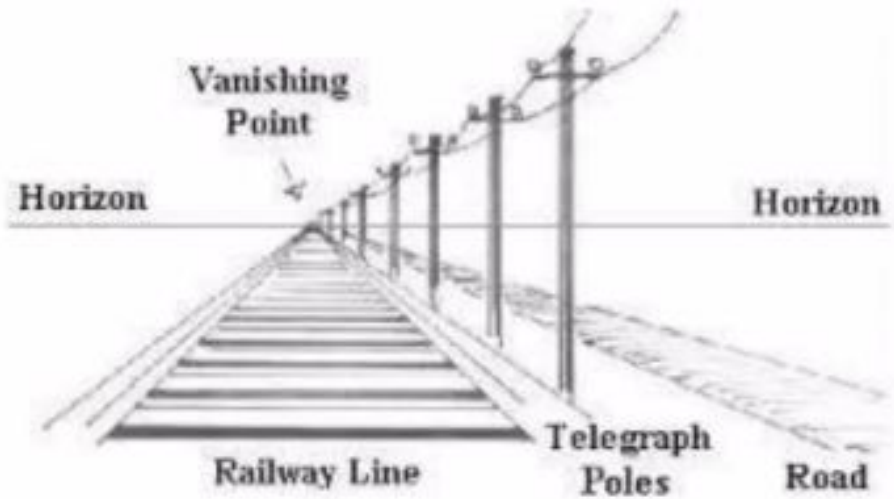
Perspective projection

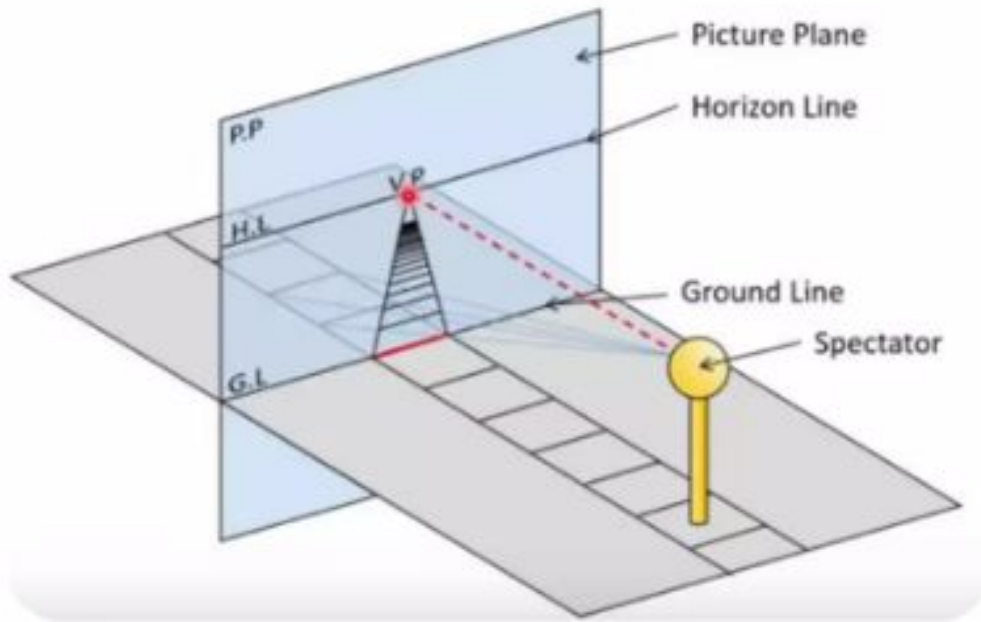
- Projection lines are drawn from the object point towards view plane.



What is Vanishing Point:

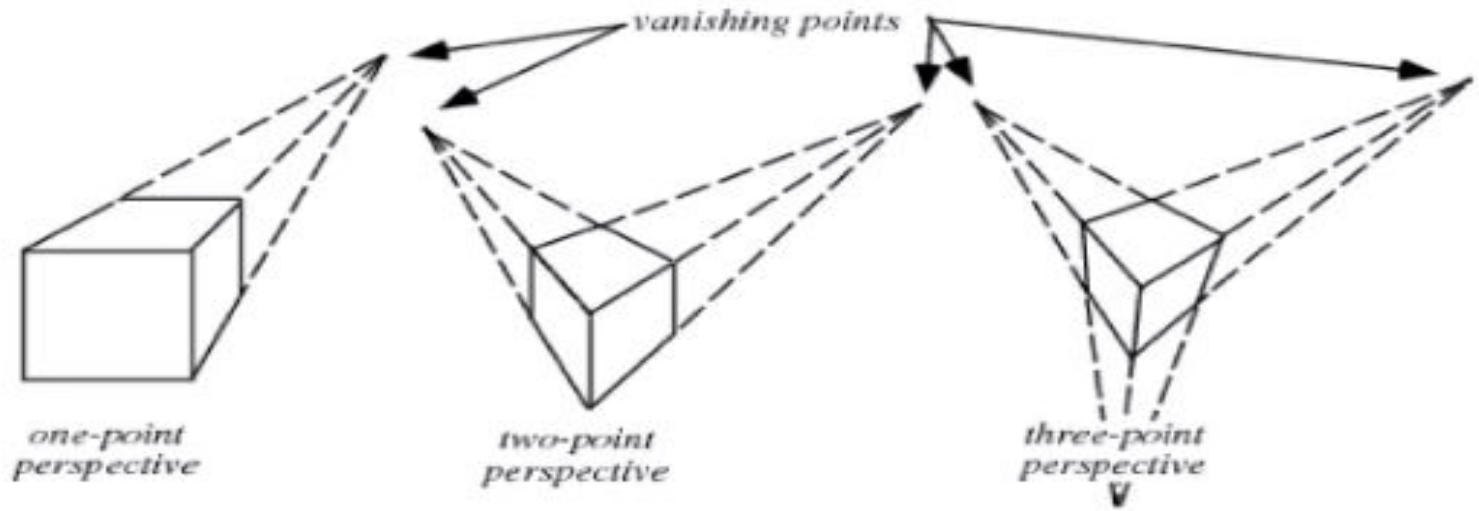
- Parallel lines that are not parallel to the viewing plane, converges to a **Vanishing Point**.





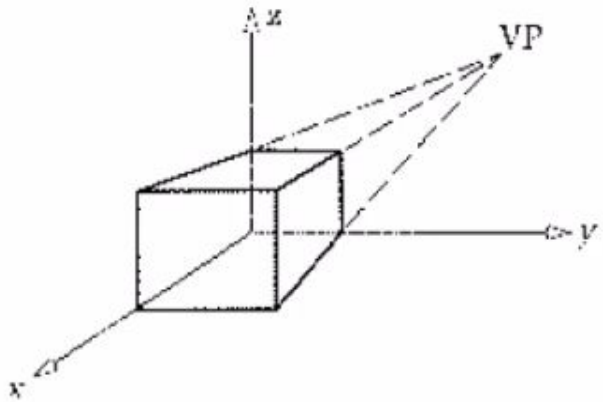
Types of Perspective Projection:

- Three types of Perspective Projection are there.
  - ✓**One Principal Vanishing Point Perspective Projection.**
  - ✓**Two Principal Vanishing Point Perspective Projection.**
  - ✓**Three Principal Vanishing Point Perspective Projection.**



One Principal Vanishing Point Perspective Projection:

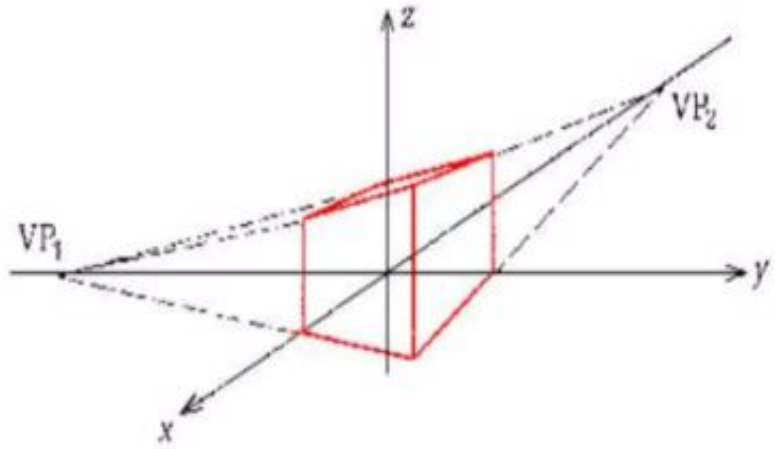
- This Perspective projection occurs when the projection plane intersect to one of the Principal axis(x, y or z).





Two Principal Vanishing Point Perspective Projection:

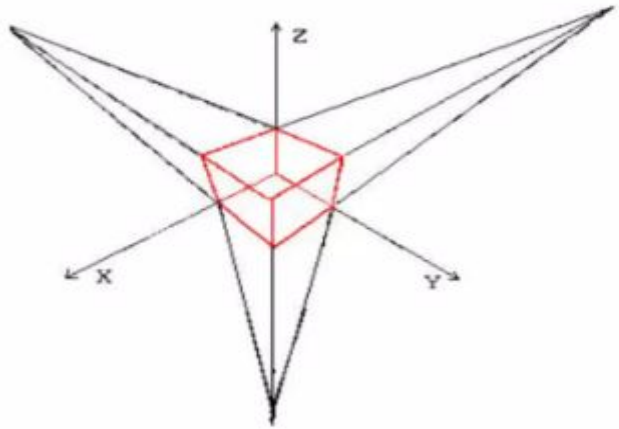
- This Perspective projection occurs when the projection plane intersects exactly any two of the Principal axis(x, y or z).





Three Principal Vanishing Point Perspective Projection:

- This Perspective projection occurs when the projection plane intersects to all the three Principal axis(x, y and z).





Monocular Depth Estimation : pinhole-camera model

$$x = f\frac{X}{Z} \qquad x = f\frac{Y}{Z}$$

where: X, Y, Z are the 3D coordinates of a point,
x, y are its image coordinates,
f is the focal length,
Z is the depth.

$$\mathrm{Z} = f\frac{X}{x}$$

However, in a single image, the real-world value X is usually unknown. Therefore, Z cannot be calculated uniquely.

For example, a small nearby object and a large faraway object may produce the same image size. This is called scale ambiguity.

When the actual object height H is known, depth can be estimated as:

$$Z = fH/h$$

where h is the object's height in the image.

Modern monocular depth-estimation systems use deep-learning models to learn a function:

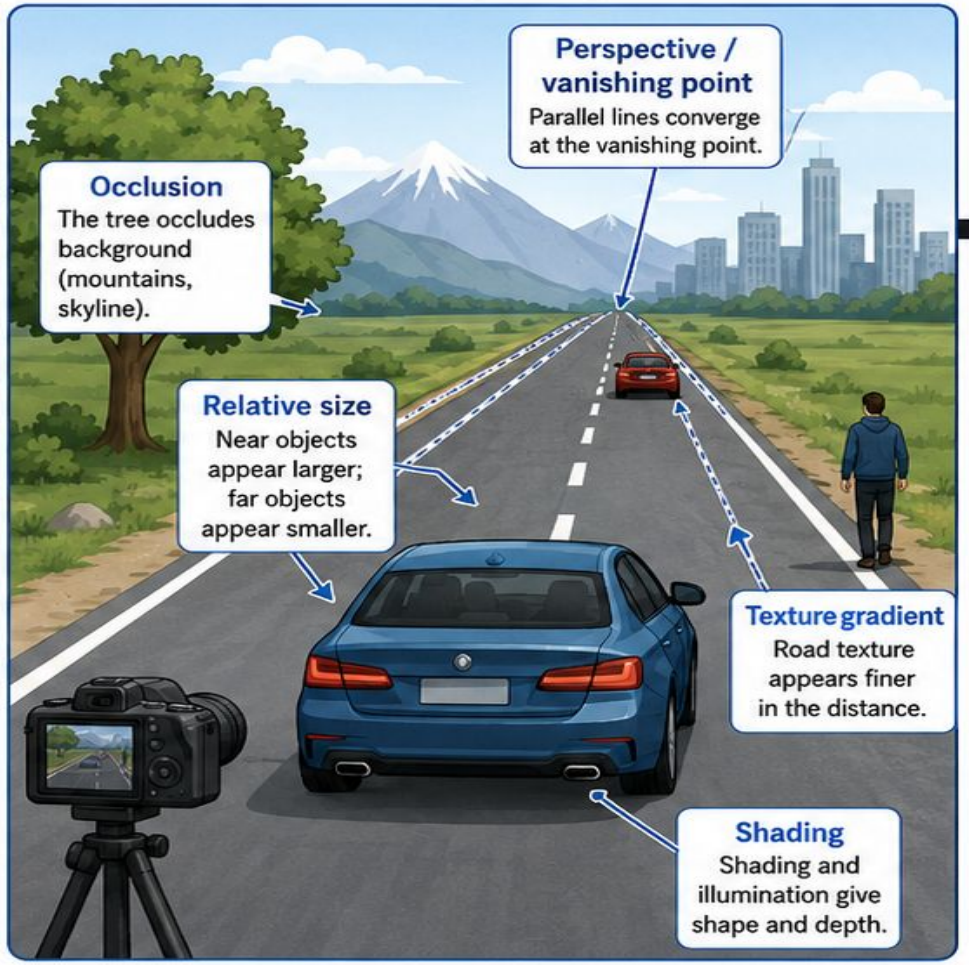
$$D\hat{} = F(I)$$

where I is the input image and
D^ is the predicted depth map.

The model learns depth using cues such as size, perspective, texture, overlap and shadows.

1 Real World Scene

Observed by a Single Camera



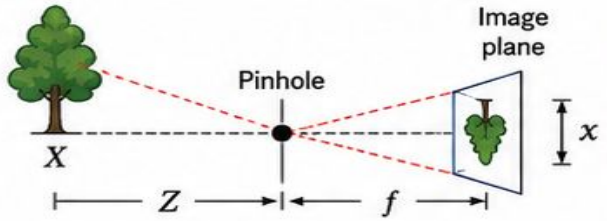
Monocular depth cues allow us to infer 3D structure from a single view.

2 Input RGB Image

Captured by the Camera



Pinhole Camera Model

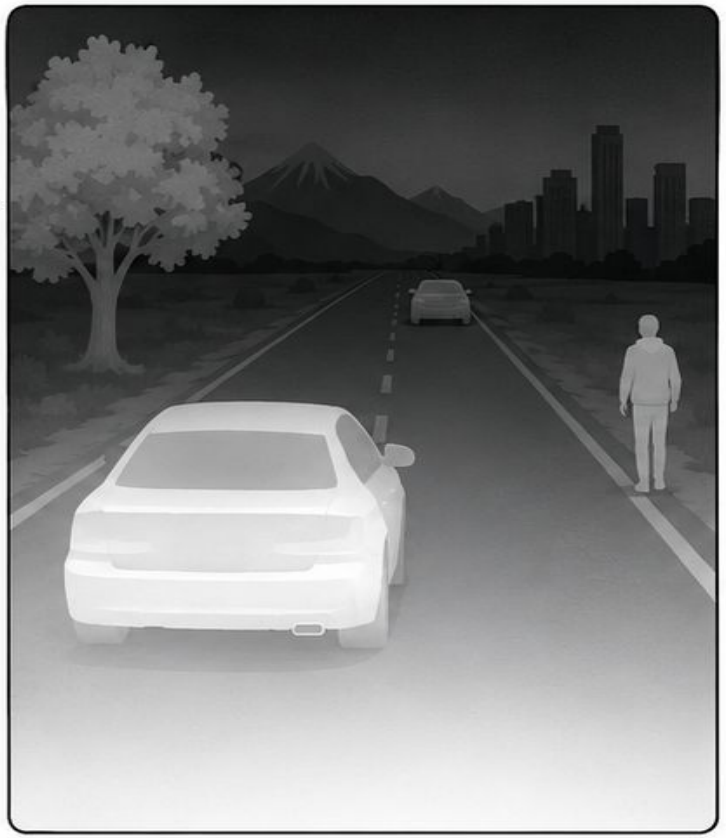


$$x = \frac{fX}{Z}$$

Depth is estimated from visual cues in a single image.

3 Output Depth Map

Predicted Depth (Grayscale)



Near = bright Far = dark

Monocular Depth Estimation: estimating scene depth from one image using geometry cues and learned visual patterns.

For example, if:

$$f = 800 \text{ pixels}, \quad H = 1.5 \text{ m}, \quad h = 300 \text{ pixels}$$

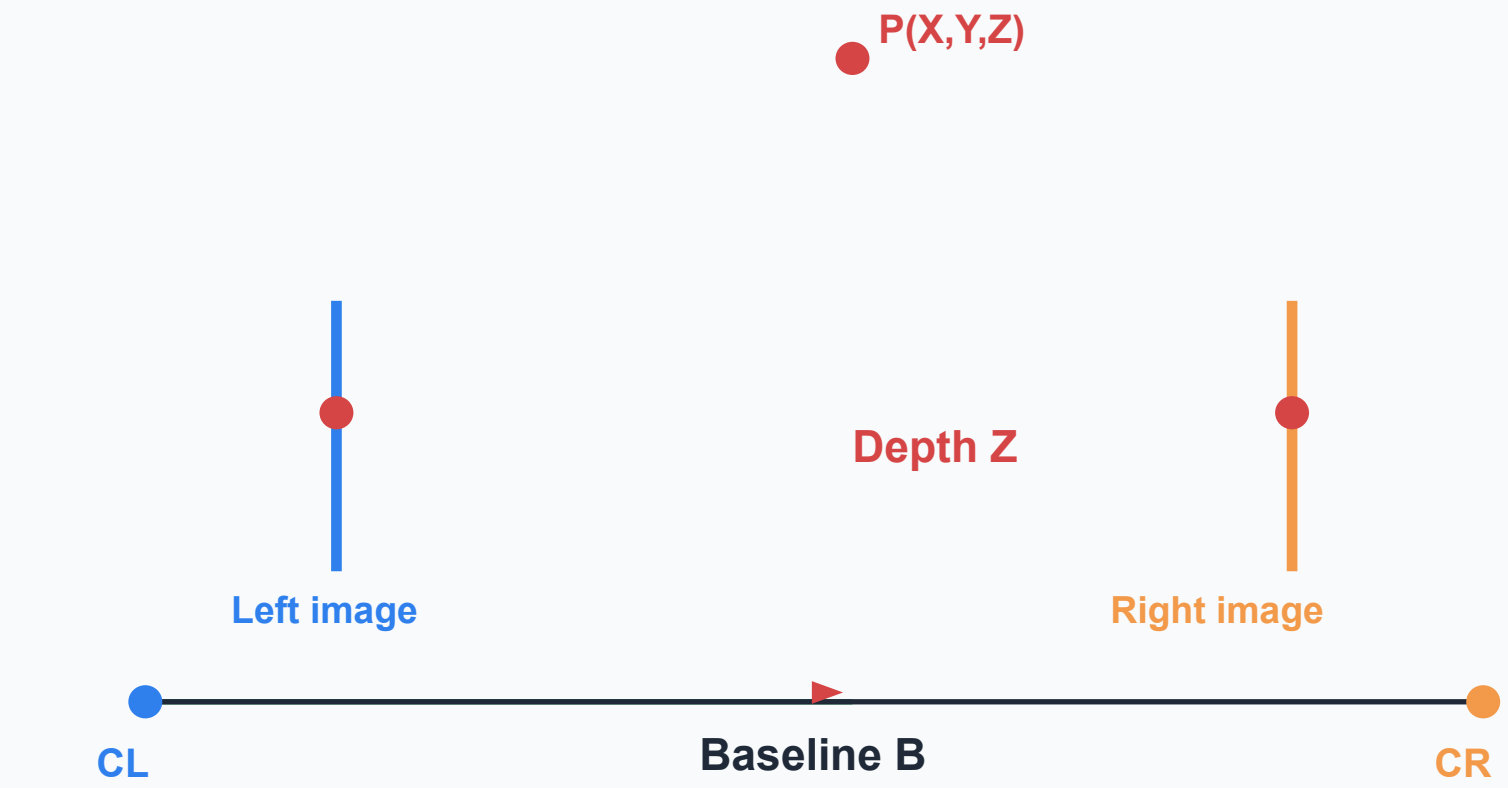
then:

$$Z = \frac{800 \times 1.5}{300} = 4 \text{ m}$$

Binocular Depth Estimation

Depth estimation using two cameras is called Stereo Depth Estimation or Binocular Stereo Vision.

Depth estimation means finding how far an object point is from the camera.

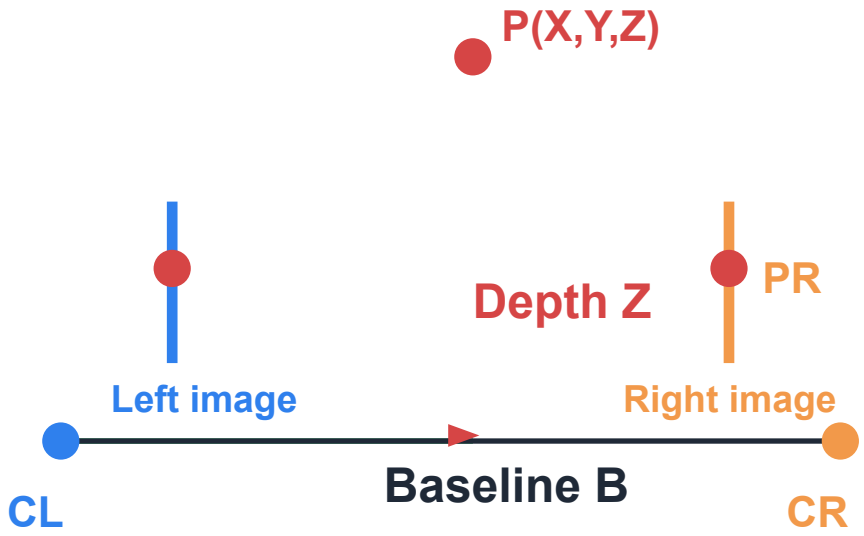


Two cameras observe the same point from different positions.

Key Idea

The same point appears shifted in two images

- Left camera captures the point at image position pL.
- Right camera captures the same point at image position pR.
- The horizontal difference between pL and pR is called disparity.
- Depth is calculated from disparity.

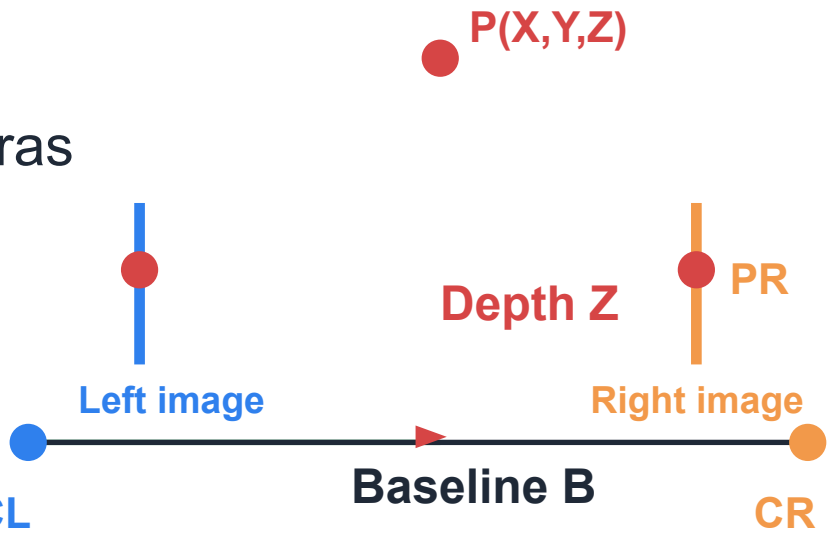


pL and pR are two image locations of the same real point P.

Important Symbols

Markers used in stereo depth

CL	centre of the left camera
CR	centre of the right camera
B	baseline: distance between the two cameras
P(X,Y,Z)	real 3D point in the scene
pL, pR	image positions of the same point P
xL, xR	horizontal pixel coordinates of pL and pR
d	disparity: xL − xR
f	focal length of the camera



Step 1: Camera Setup

Rectified stereo vision assumption

- Two cameras are parallel and horizontally separated.
- Both cameras have focal length f.
- The distance between cameras is baseline B.
- After rectification, matching points lie on the same horizontal row.

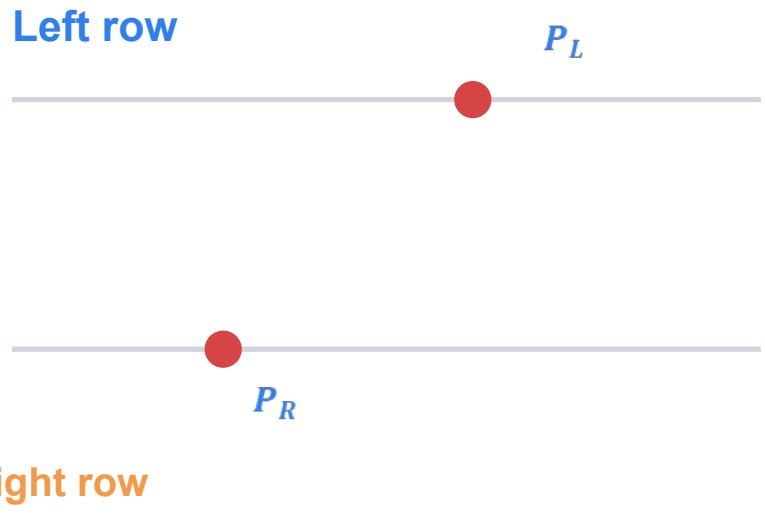
Step 2: Find Matching Points

Match the same scene point in left and right images

The real point P is projected into two image points:

$$P_L = (x_L, y_L) \; in \; the \; left \; image$$

$$P_R = (x_R, y_R) \; in \; the \; right \; image$$



For rectified stereo:

$$x_L = y_R$$

So the search becomes easier: match along the same row.

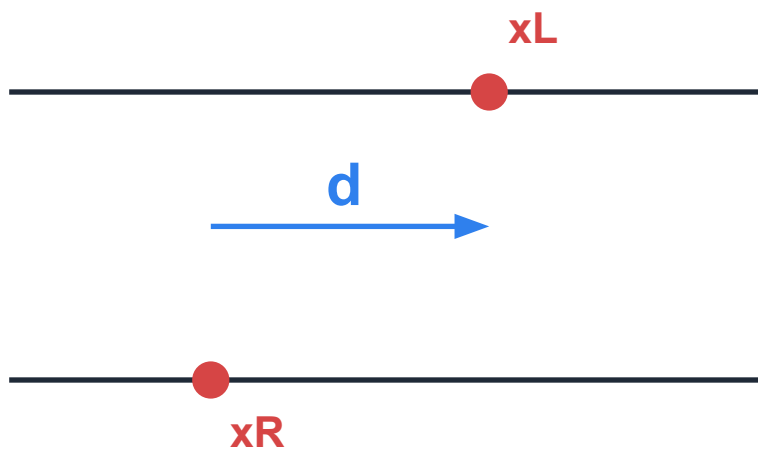
Step 3: Calculate Disparity

Disparity is horizontal image shift

Disparity formula:

$$d = x_L - x_R$$

where xL and xR are measured in pixels.



- Large disparity means the point shifted a lot.
- Small disparity means the point shifted very little.
- This shift contains depth information.

Step 4: Pinhole Projection for Left Camera

Projection equation for the left camera

Let the real 3D point be P(X, Y, Z).

Using the pinhole camera model:

$$x_L = f\frac{X}{Z}$$

This says: image position depends on real position X, focal length f, and depth Z.

Rearrange it: $$X = Z\frac{x_L}{f}$$

Step 5: Projection for Right Camera

Right camera is shifted by B

The right camera is B units to the right of the left camera.

So the horizontal coordinate seen by the right camera becomes X − B.

Right camera projection:

$$x_R = f\frac{X - B}{Z}$$

Now expand:

$$x_R = \frac{fX}{Z} - \frac{fB}{Z}$$

$$x_R = x_L - \frac{fB}{Z}$$

$$d = x_L - x_R$$

$$d = x_L - x_L + \frac{fB}{Z}$$

$$d = \frac{fB}{Z}$$

$$Z = \frac{fB}{d}$$

This is the formula of stereo depth estimation.

Numerical Example

Calculate disparity first, then depth

Given:

f = 800 pixels

B = 0.10 m

xL = 500 pixels

xR = 460 pixels

Step 1: d = xL − xR

d = 500 − 460 = 40 pixels

Step 2: Z = fB/d

Z = (800 × 0.10) / 40

Z = 80 / 40

Z = 2 m

The point is 2 metres away from the camera.

Activa

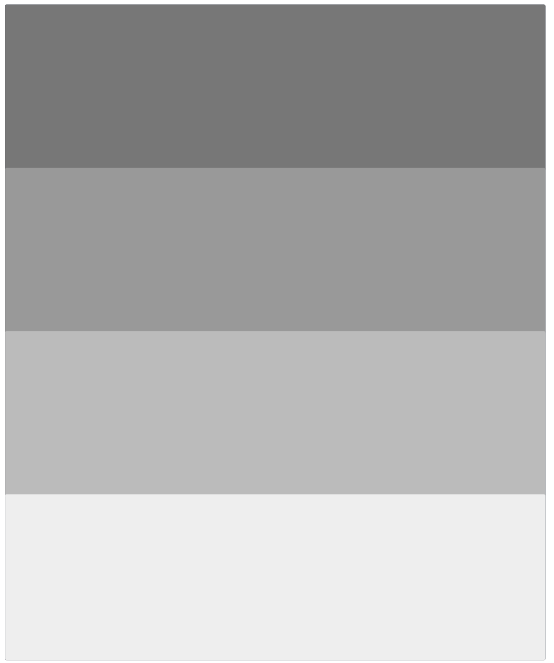
From Disparity Map to Depth Map

Repeat the same calculation for every pixel

For one point: $Z = \frac{fB}{d}$

For every pixel: $Z(u, v) = \frac{fB}{d(u,v)}$

- D(u,v) is the disparity value at pixel (u,v).
- Applying the formula to all pixels gives a depth map.
- A depth map tells how far each visible scene point is.



Depth map

dark = far bright = near

Camera Geometry

Camera geometry explains how a real 3D point in the world is projected onto a 2D image captured by a camera.

A camera converts:

$$3D\ point\ P(X, Y, Z)\ into\ 2D\ image\ point\ p(u, v)$$

Camera centre C: The camera centre is the point from which the scene is observed.

Optical axis: The optical axis is an imaginary straight line passing from the camera centre through the centre of the image.

Image plane: The image plane is the surface where the 2D image is formed.

A point on the image is represented using pixel coordinates:

$$p = (u, v)\ where:$$

$$u = horizontal\ pixel\ position$$

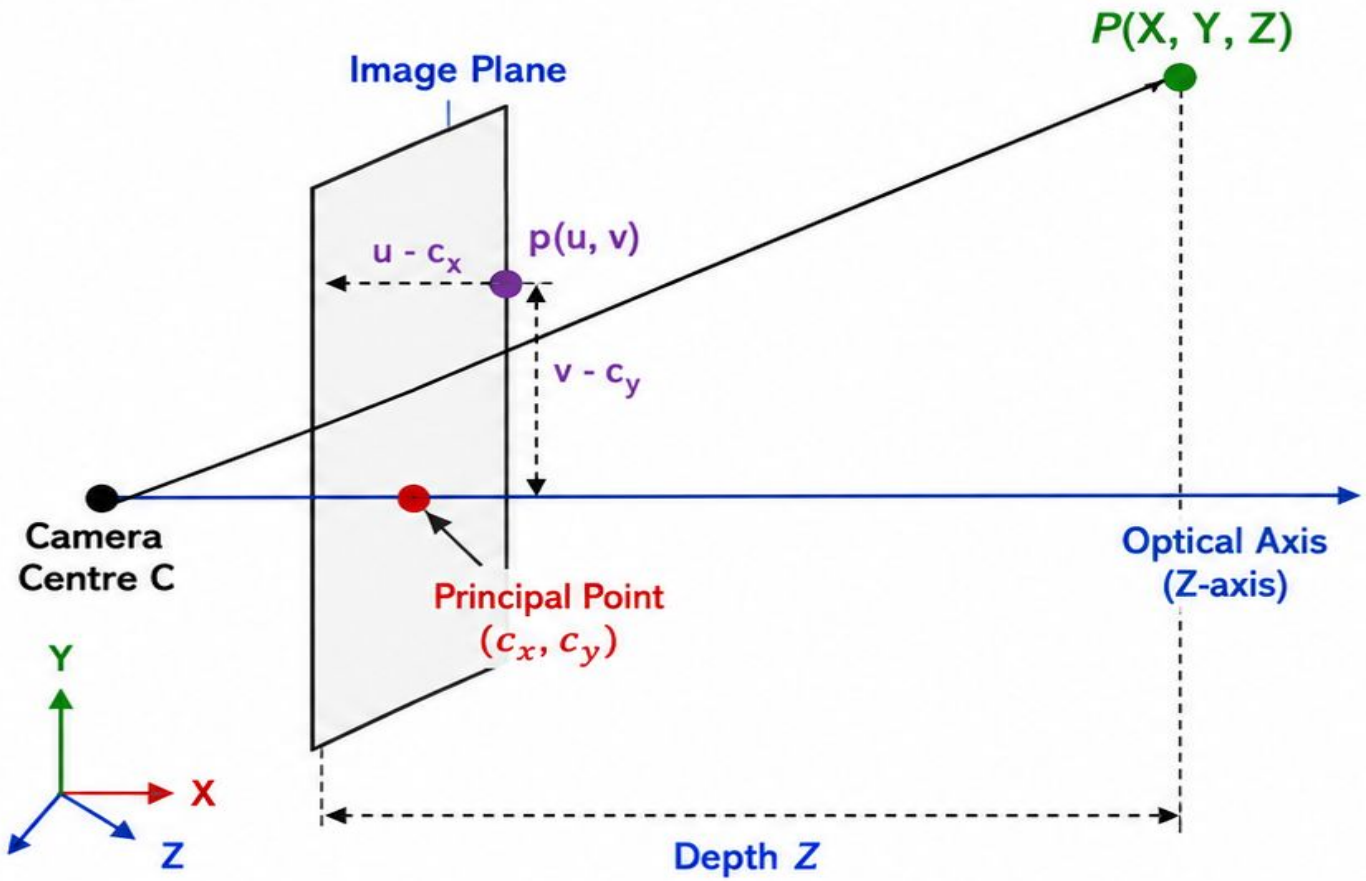
$$v = vertical\ pixel\ position$$

Focal length f: The focal length is the distance between the camera centre and the image plane.

Principal point: The point where the optical axis meets the image plane is called the **principal point**.

It is represented as: (cx,cy)

1. Pinhole Camera Model



2. Projection Equations

$$u = c_x + \frac{f_x X}{Z}$$

$$v = c_y + \frac{f_y Y}{Z}$$

Where:

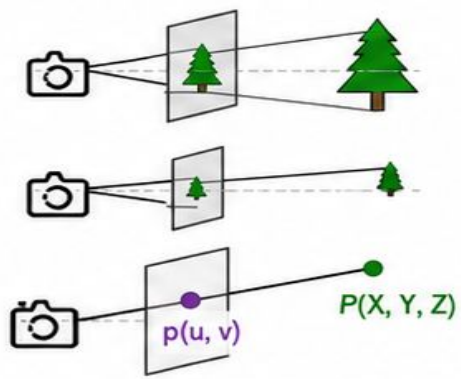
- (u, v) = image pixel coordinates
- (X, Y, Z) = 3D camera coordinates
- f_x, f_y = focal lengths in pixels
- (c_x, c_y) = principal point

As depth Z increases, the image projection becomes smaller.

$$X = \frac{(u - c_x)\, Z}{f_x} \qquad Y = \frac{(v - c_y)\, Z}{f_y}$$

3. Intuition

- Near object -> appears larger in the image
- Far object -> appears smaller in the image
- Camera geometry links 3D world points to 2D image points



Intrinsic camera parameters

The internal camera parameters are represented by the intrinsic matrix:

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

It contains:

- focal lengths f_x, f_y
- principal point c_x, c_y

The projection equation is:

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

Here, λ is a scale value related to depth.

Extrinsic camera parameters

Extrinsic parameters describe the camera position and orientation in the world.

They are:

- R: rotation matrix
- t: translation vector

The complete camera projection equation is:

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} R & t \end{bmatrix} \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$$

The extrinsic parameters first convert a world point into camera coordinates. The intrinsic parameters then project it onto the image.

Homography

A **homography** is a transformation that maps points from one image plane to another image plane.

It is commonly used when:

- the scene is approximately planar, such as a wall, road, page, or board
- or the camera rotates without changing its position

A homography can relate the same point in two images.

$$p' \sim Hp$$

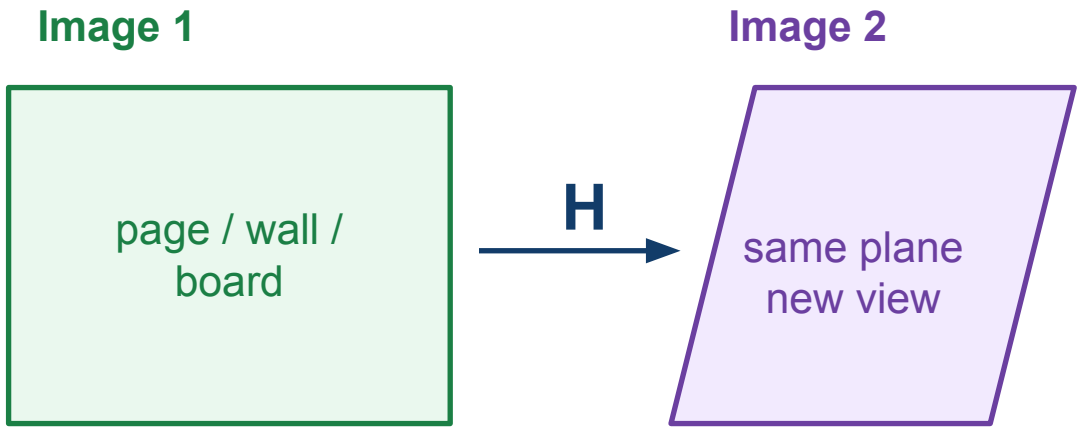
where:

p = point in the first image

p′ = corresponding point in the second image

H = 3×3 homography matrix

~ means equality up to a scale factor



Point Coordinates and the Matrix H

Homography uses homogeneous coordinates

Image point in homogeneous form:

$$p = [\, x,\ y,\ 1 \,]^T$$

Corresponding point in another image

$$p = [\, x',\ y',\ 1 \,]^T$$

Homography matrix:

$$H = \begin{bmatrix} h11 & h12 & h13 \\ h21 & h22 & h23 \\ h31 & h32 & h33 \end{bmatrix}$$

After multiplication:

$$[\, x', y', w' \,]^T = H\, [\, x, y, 1 \,]^T$$

Convert back to pixel coordinates:

$$X' = x' / w'$$
$$Y' = y' / w'$$

Why w′?
Homogeneous coordinates allow perspective effects like vanishing points and trapezium-like distortion.

Why is homography needed?

Homography is used to transform one view of a plane into another view.

Applications include:

- image stitching and panorama creation
- document scanning
- perspective correction
- image alignment
- augmented reality
- planar object tracking
- road-view transformation
- camera calibration

Uses of Homography

1. **Image Stitching and Panorama Creation**

Homography aligns overlapping parts of two or more images.

Example:A person captures three photos of a landscape by rotating the camera.

Homography matches common points such as trees and buildings, then combines the images into one wide panorama.p′~Hp2.

2. **Document Scanning**

Homography converts a tilted photograph of a document into a flat, front-facing view.

Example: A mobile phone captures a page from an angle.

The rectangular page looks like a trapezium. Homography transforms its four corners into a proper rectangle.

3. **Perspective Correction**

Homography removes perspective distortion from planar objects.

Example: A classroom board photographed from the side appears narrower at one end. Homography changes it into a straight rectangular front view.

6. Planar Object Tracking

Homography tracks the movement and shape of a flat object across video frames.

Example:A system tracks a book cover in a video. Even when the book rotates or tilts, homography estimates its new position and boundary.

7. **Road-View Transformation**

Homography converts the normal camera view of a road into a top view, also called a bird’s-eye view.

Example:A camera mounted on a car captures the road in perspective. Homography transforms the road plane so that lane lines appear parallel and distances are easier to measure. This is often called Inverse Perspective Mapping.

2. What is epipolar geometry?

Epipolar geometry describes the relationship between two cameras observing the same 3D scene.

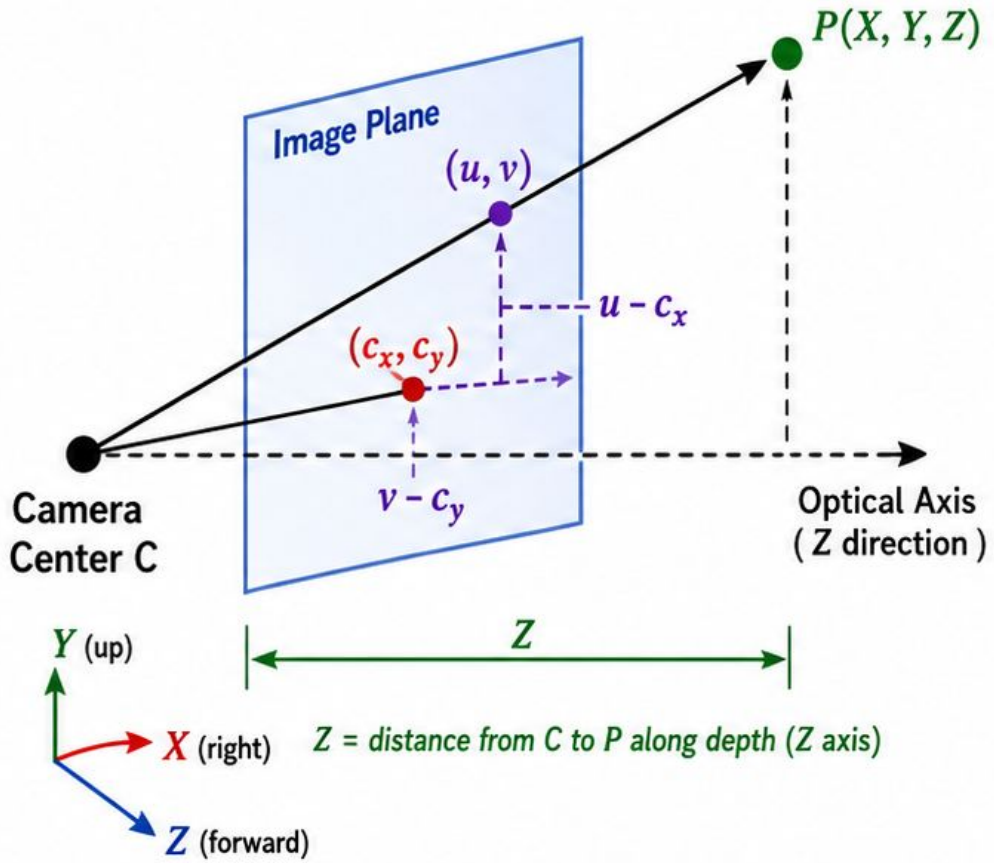
Its main components are:

- camera centres
- baseline
- epipolar plane
- epipoles
- epipolar lines
- corresponding image points

Recovering Complete 3D Coordinates from a Depth Pixel

From image pixel (u, v) and depth Z to 3D point $P(X, Y, Z)$

1. Camera Geometry



2. Step-by-step equations

1 Depth from stereo disparity:

$$Z = \frac{f_x B}{d}$$

2 Horizontal image projection:

$$u = c_x + f_x X$$

$$u - c_x = f_x X / Z$$

$$X = ((u - c_x) Z / f_x$$

3 Vertical image projection:

$$v = c_y + f_y Y$$

$$v - c_y = f_y Y / Z$$

$$Y = ((v - c_y) Z / f_y$$

4 Final 3D point:

$$P = (X, Y, Z)$$

3. Numerical Example

$u = 420,\ v = 280$

$c_x = 320,\ c_y = 240$

$f_x = 800,\ f_y = 800$

$B = 0.1$ m

$d = 40$ pixels

Z = (800 x 0.1) / 40 = 2.0 m

X = ((420 - 320) x 2.0) / 800 = 0.25 m

Y = ((280 - 240) x 2.0) / 800 = 0.10 m

P = (0.25, 0.10, 2.0) m

The point is 0.25 m to the right, 0.10 m vertically, and 2.0 m in front of the camera.

Meaning of symbols

(u, v) =	(c_x, c_y) =	f_x, f_y =	B =	d =	$P(X, Y, Z)$ =
pixel coordinates	principal point	focal lengths in pixels	baseline	disparity	3D world point

3. Camera Geometry

Internal camera parameters + external pose

$$s\,x = K\,[\,R \mid t\,]\,X$$

Intrinsic K
inside the camera

→

Extrinsic R, t
position + orientation

→

Projection matrix P
P = K [R | t]

- K contains focal length, principal point and skew.
- R rotates world coordinates into camera coordinates.
- t translates world origin into camera coordinates.
- Together they map a 3-D point to a pixel.

Intrinsic Matrix K

The camera’s internal measurement system

$$K = \begin{bmatrix} fx & s & cx \\ 0 & fy & cy \\ 0 & 0 & 1 \end{bmatrix}$$

fx, fy
focal length in pixels

cx, cy
principal point

s
skew between axes

distortion
lens bending effects

$$u = fx \cdot X/Z + cx$$
$$v = fy \cdot Y/Z + cy$$

4. Binocular Stereopsis

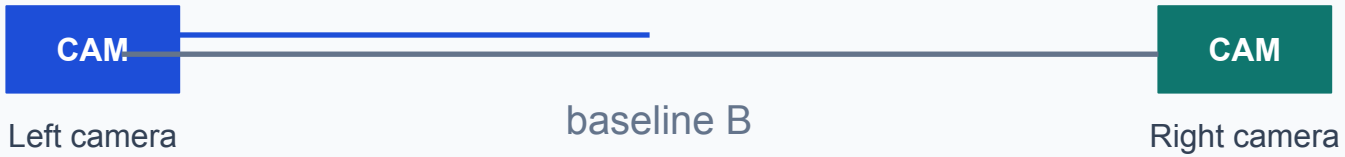
Two eyes / two cameras create depth through disparity

$$d = xL - xR$$



$$Z = fB / d$$

Different image positions → disparity → depth



Stereo Depth: Worked Example

From pixel disparity to real distance

Left image
xL = 420 px



Right image
xR = 380 px

$$d = 420 - 380 = 40 \text{ px}$$

$$Z = (800 \times 0.12) / 40 = 2.4 \text{ m}$$

Large disparity ⇒ object is near
Small disparity ⇒ object is far

Stereo Correspondence Is the Hard Part

Which right-image pixel matches the left-image pixel?

Textureless surface

Repeated patterns

Occlusion

Reflection

Low light

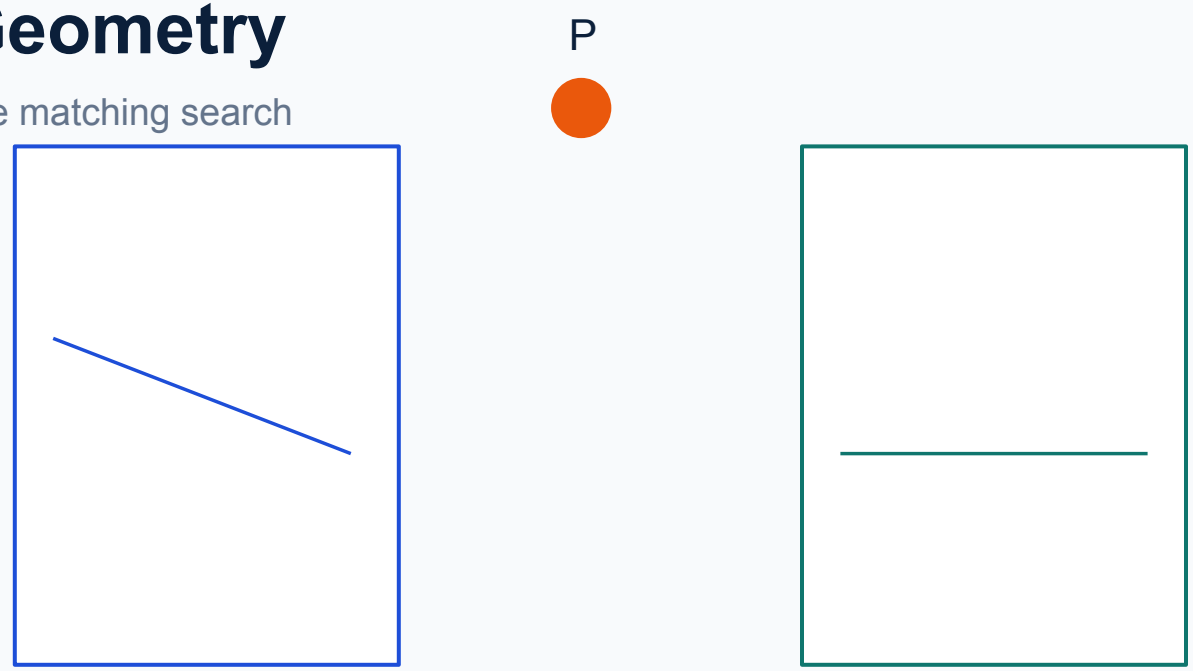
Motion blur

Depth error grows rapidly when disparity is small:

$$\Delta Z \approx -(fB / d^2)\,\Delta d$$

5. Epipolar Geometry

Geometry that restricts the matching search



Epipolar line = possible location of corresponding point



Fundamental Matrix F

Relates corresponding points in two uncalibrated images

$$x'^{T} F x = 0$$



- F is a 3 × 3 matrix.
- It is defined up to scale.
- It has rank 2 and 7 degrees of freedom.
- It is estimated from point correspondences.

Essential Matrix E

Epipolar geometry when camera calibration is known

$$E = [t]_\times R$$

$$F = K_2^{-T} E K_1^{-1}$$

R
relative rotation

→

t
translation direction

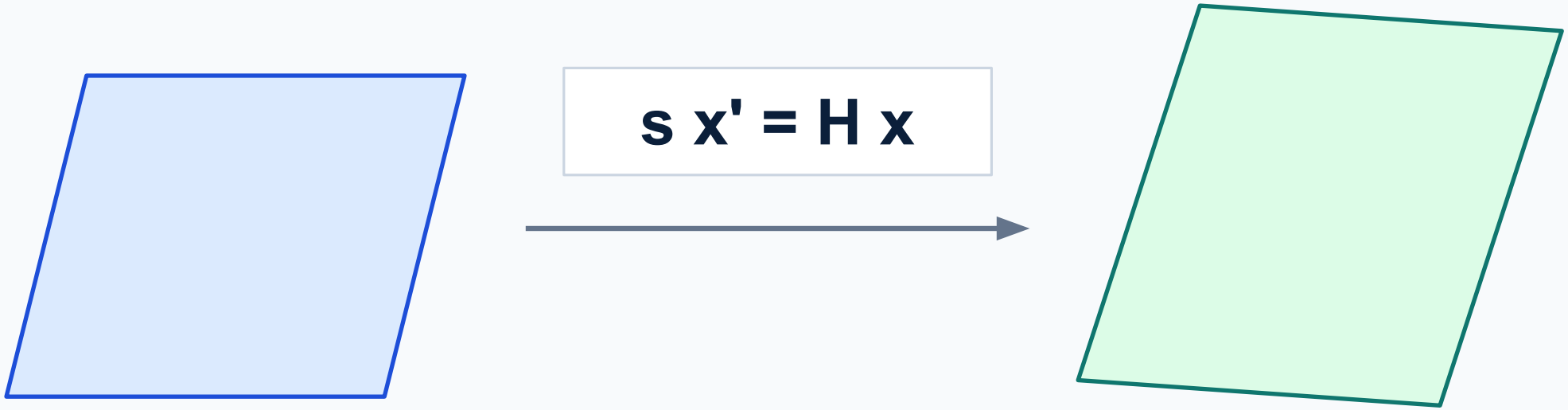
→

E
calibrated geometry

- Use E when intrinsic matrices are known.
- E helps recover relative camera pose.
- Translation magnitude is still unknown without scale.

6. Homography

A projective mapping between two views of a plane



- H is a 3 × 3 matrix with 8 independent parameters.
- Minimum 4 non-collinear correspondences are required.
- Works exactly for planar scenes or pure camera rotation.

Homography in Practice

Correcting and aligning planar views

Document scanning

Panorama stitching

AR marker tracking

Bird’s-eye road view

Point transformation:

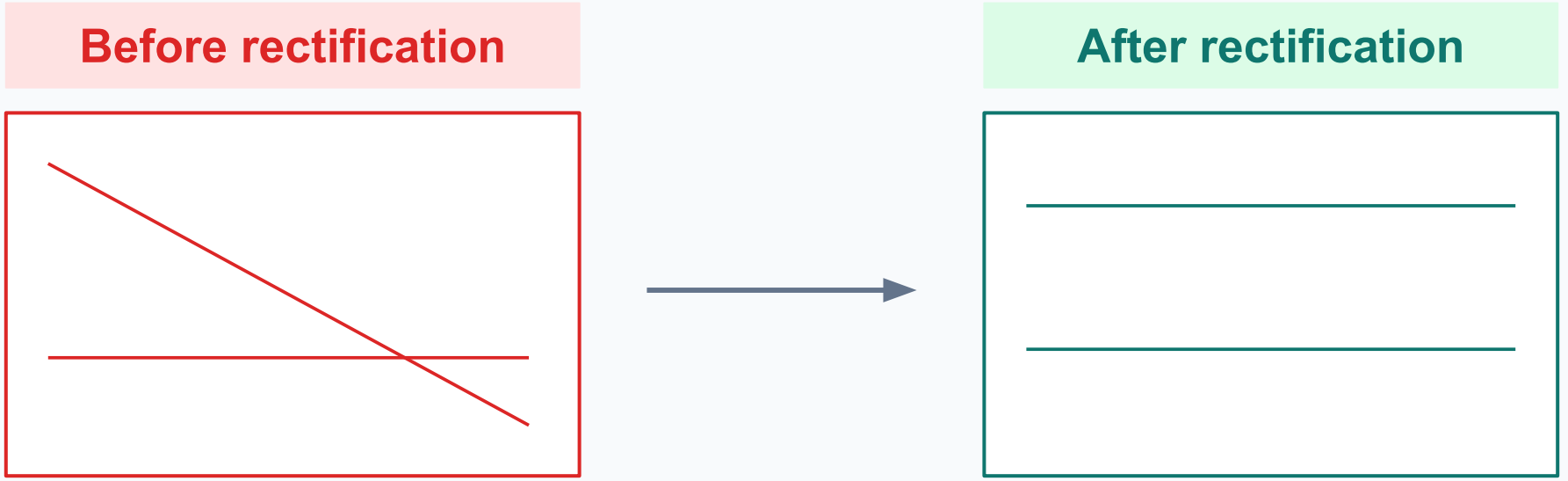
$$[u, v, w]^T = H\,[x, y, 1]^T$$

$$x' = u/w, \quad y' = v/w$$

Warning: a single homography cannot correctly model a general 3-D scene with camera translation.

7. Image Rectification

Make epipolar lines horizontal and aligned



Before: yL ≠ yR
After: yL = yR

Rectification Pipeline

Calibrated stereo workflow

Calibrate cameras

Estimate R and t

Compute rectifying rotations

Warp images

Search along rows

Output: rectified image pair + valid image region

8. Direct Linear Transform (DLT)

Estimate projective models from point correspondences

$$A\,h = 0$$

Collect matches → **Normalize points** → **Build matrix A** → **Solve using SVD** → **Reshape vector** → **Evaluate error**

- DLT is used for homography, camera matrix and triangulation.
- It is simple and fast, but assumes correspondences are correct.
- Point normalization improves numerical stability.

DLT for Homography

Each correspondence gives two linear equations

$$x \leftrightarrow x', \quad x' \sim Hx$$

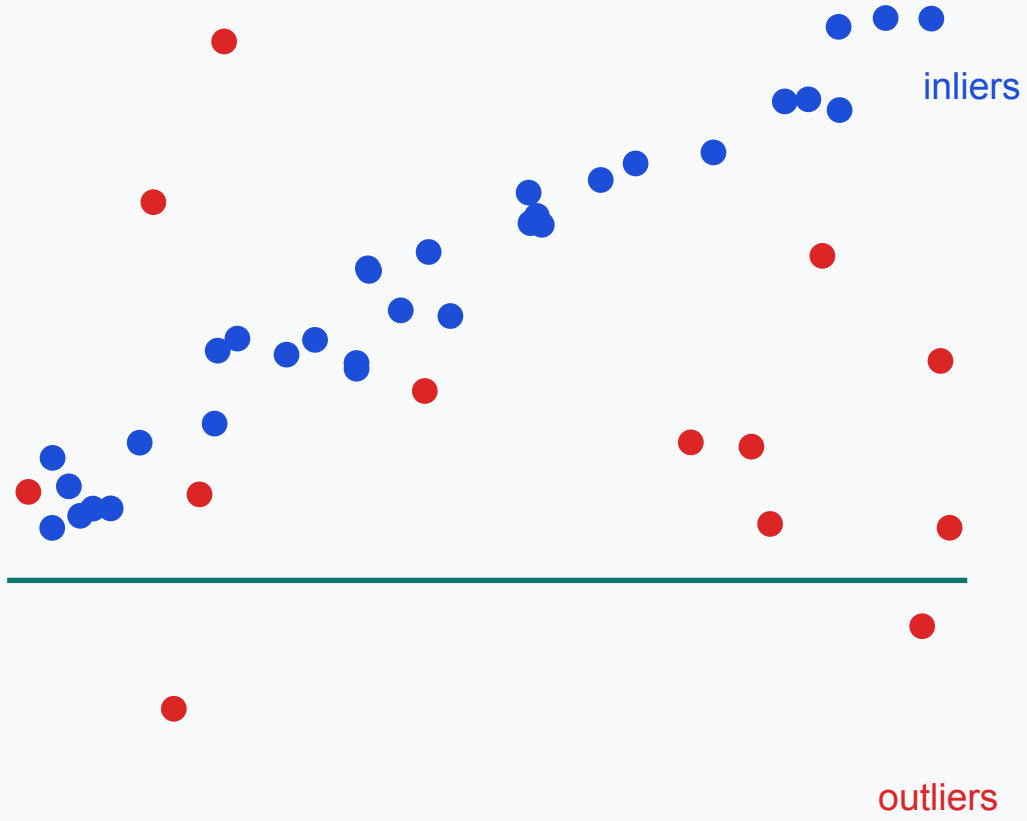
$$\begin{bmatrix} -x & -y & -1 & 0 & 0 & 0 & ux & uy & u \\ 0 & 0 & 0 & -x & -y & -1 & vx & vy & v \end{bmatrix} h = 0$$

Minimum for homography: 4 non-collinear matches

Final step: choose the right singular vector corresponding to the smallest singular value.

9. RANSAC

Robust estimation when feature matches contain outliers



- Randomly sample minimal subset.
- Fit a candidate model.
- Count inliers using an error threshold.
- Keep the model with maximum consensus.

RANSAC Algorithm

A practical recipe for robust geometry

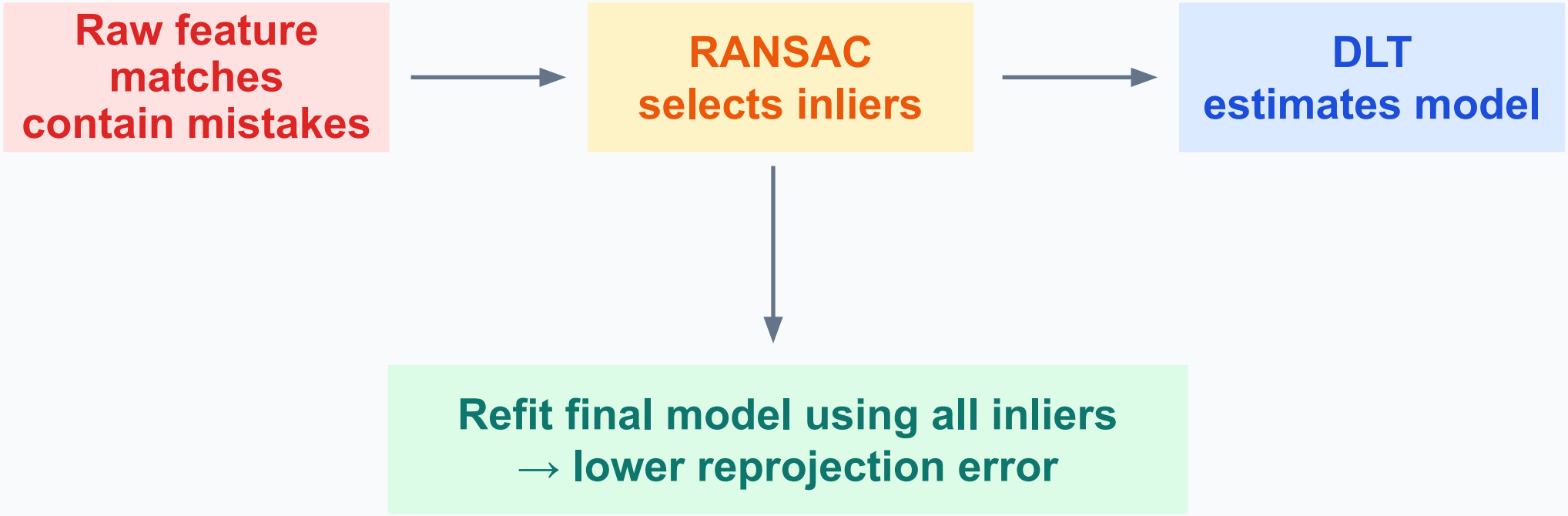
1. Sample minimal points
2. Fit model
3. Measure error
4. Count inliers
5. Repeat and select best

$$N = \log(1 - p) / \log(1 - w^s)$$

- p = desired confidence w = inlier ratio s = sample size

DLT + RANSAC

The standard practical combination



10. 3-D Reconstruction Framework

From many images to a 3-D model



Goal: estimate both camera motion and scene structure.